On Mathews Correlation Coefficient and Improved Distance Map Loss for Automatic Glacier Calving Front Segmentation in SAR Imagery

Amirabbas Davari, Saahil Islam, Thorsten Seehaus, Andreas Hartmann, Matthias Braun, Andreas Maier, *Senior Member, IEEE*, and Vincent Christlein

***Abstract*— The vast majority of the outlet glaciers and ice streams of the polar ice sheets end in the ocean. Ice mass loss via calving of the glaciers into the ocean has increased over the last few decades. Information on the temporal variability of the calving front position (CFP) provides fundamental information on the state of the glacier and ice stream, which can be exploited as calibration and validation data to enhance ice dynamics modeling. To identify the CFP automatically, deep neural network-based semantic segmentation pipelines can be used to delineate the acquired synthetic aperture radar (SAR) imagery. However, the extreme class imbalance is highly challenging for the accurate calving front segmentation in these images. Therefore, we propose the use of the Mathews correlation coefficient (MCC) as an early stopping criterion because of its symmetrical properties and its invariance toward class imbalance. Moreover, we propose an improvement to the distance map-based binary cross-entropy (BCE) loss function. The distance map adds context to the loss function about the important regions for segmentation and helps to account for the imbalanced data. Using MCC as early stopping demonstrates an average 2.16% dice coefficient improvement for the segmentation of the slightly imbalanced ice/nonice regions and 17.43% dice coefficient improvement for segmenting the highly imbalanced CFPs, compared to the commonly used BCE. The modified distance map loss further improves the calving front segmentation performance by another 1.6%. These results are encouraging as they support the effectiveness of the proposed methods for segmentation problems suffering from extreme class imbalances.**

***Index Terms*— Class imbalance, improved distance map loss, Mathews correlation coefficient (MCC), semantic segmentation.**

I. Introduction

THE mass loss of the polar ice sheets and other calving glaciers has accelerated in the last two decades. This is one of the main causes of global sea-level rise. Currently, the sea-level contribution from thermal expansion is exceeded by the contribution from ice sheets, glaciers, and ice caps [1]. The retreat of glacier fronts can initiate feedback cycles. For example, the retreat from the pinning points, such as basal or lateral bedrock formation or subaquatic moraines, can reduce the buttressing forces, which further destabilizes the glaciers' ice discharge into the ocean [2]. A significant frontal retreat can further cause a retreat of the grounding zone leading to further destabilization of the glacier flow [3]. The calving front is defined as the region where the glacial ice breaks off into the ocean. Hence, the calving front position (CFP) is an important variable for monitoring glacier changes but also for ice dynamics modeling.

The detection of the calving front location is commonly carried out using optical imagery or synthetic aperture radar (SAR) imagery. Cloud cover and polar night affect the optical imagery and the data coverage is strongly limited. Conversely, SAR acquisitions are unaffected by such situations and can be carried out all year round, making SAR imagery more advantageous for monitoring purposes in polar regions. The common approach to detect CFPs has been visual analysis and manual delineation on the obtained SAR images [4], [5]. However, manual digitization is tedious, subjective, expensive, and time-consuming. Moreover, sea ice and calved-off icebergs in the oceans make it often difficult to reliably differentiate between ice mass and these chunks of floating ice, making manual delineation subjective. This leads to a lack of temporally updated data on glacier area, which is an essential climate variable (ECV) product [6]. Automatic detection of CFPs would close this monitoring gap. In the past, various algorithms of automatic and semiautomatic detection of the CFPs had been proposed. Edge detection algorithms, such as Roberts edge extractor, have been applied to ERS-1 SAR images by Sohn and Jezek [7]. Seale *et al.* [8] have exploited the Sobel filter and brightness profiling to detect the CFPs on daily MODIS images. A detailed overview of various automatic and semiautomatic approaches on this application is provided by Baumhoer *et al.* [5].

Deep learning-based methods, such as convolutional neural networks (CNNs) [9], are a better alternative to the traditional automatic delineation of CFPs. Over the past two decades, CNNs have been seen to outperform the abovementioned

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manually designed filter-based methods and many such traditional computer vision techniques in the field of image processing [10]. The CNN architecture (the type and number of layers and their order) is chosen depending on the application and the training data. The parameters in a CNN are fine-tuned based on optimizing a predefined objective function, also referred to as cost function or loss function.

Deep convolutional networks can be used on SAR images to automatically classify the image pixels, belonging to semantically different regions, into distinct classes. Such a task is referred to as semantic segmentation. Architectures, such as SegNet [11] and U-Net [12], are commonly used for semantic segmentation applications [9]. The use of the U-Net architecture for image segmentation to automatically detect calving fronts was first exploited by Mohajerani *et al.* [13] on Landsat-5, 7, and 8 images. Zhang *et al.* [14] have used U-Net on higher resolution TerraSAR-X (TSX) images of Jakobshavn Isbræ glaciers to separate the whole ice mélange regions from nonice mélange regions, which are subsequently postprocessed to manually delineate the calving fronts. A similar methodology of predicting calving front locations is carried out by Cheng *et al.* [15] where they have used a so-called calving front machine neural network (based on Xception blocks and atrous spatial pyramid pooling) over both optical and SAR images of 66 Greenlandic glaciers. Moreover, a comparable application of CNNs on SAR imagery is the detection of sea ice, which is also an emerging area of work [16]–[19].

Despite being powerful learning tools, the high number of parameters in deep neural networks (DNNs) comes with the risk of overfitting to the training data [20], i. e., it "memorizes" the nonpredictive features of the training data instead of "learning" its underlying distribution.

Early stopping suggests stopping training based on the error on independent validation data to maintain high performance on new unseen data from the same distribution. Since the training gets terminated based on the error on the validation data, it is crucial to have the right performance metric as a stopping criterion. The traditional method of early stopping comprises monitoring the loss [21]. However, the significance of the performance metric increases in cases of class-imbalanced data [22]. An imbalanced dataset comprises classes with an unequal number of class samples. The majority of performance metrics are challenged in the presence of severely imbalanced data. For example, accuracy is not an appropriate metric in applications with class imbalance as incorrect classifications of the minority class could be overlooked. Conversely, other metrics such as f1-score greatly diminishes the effect of class imbalance. Mathews correlation coefficient (MCC) theoretically has a higher tolerance over class imbalance than f1-score and confusion entropy [23], [24] and was, for example, used to compare between a vast range of classifiers for evaluation of models on microarray gene expression and genotyping data [25]. In this article, we are suggesting the use of early stopping to prevent overfitting and using MCC instead of the classical binary cross-entropy (BCE) loss as a stopping criterion.

While early stopping handles overfitting with the right performance metric, it is also crucial to objectively handle the learning of the network on the imbalanced dataset. The imbalanced dataset introduces a bias toward the majority class during the training of the network, while the minority class is the point of interest in many real-world problems. Common solutions are, for example, data resampling [26], cost-sensitive learning, one class learning, or ensemble learning methods [27]. While these methods could be effective, they do not guarantee to be so in all kinds of tasks. For example, oversampling undergoes a risk of early overfitting because the network uses many samples more than once, whereas undersampling reduces the size of the training data. Artificially balancing the class distributions was not very successful in some tasks [28] and many ensemble-based methods are resource-demanding and time-consuming [27].

Another way of cost-sensitive learning is to weight the minority class higher to punish misclassification. In imbalanced segmentation tasks, the loss function can also be trained to penalize more the contour of the segmentation masks than the regions. Kervadec *et al.* [29] used such a boundary-based loss function for medical image segmentation. They argued that region integrals of the segmentation masks highly vary over the classes and instead trained on the space of contours. Another distance-based loss approach was employed to segment knee bones in 3-D MRI [30]. However, they employed the idea of a distance map-based loss because most errors in segmentation are found to be at the edges of the segmented masks. A distance map-based loss was also used by Adhikari and Kim [31] to detect forest trails in optical images of forests, which covers only small regions in the images. Therefore, the trails were weighted higher in the loss function, while the other regions were weighted less based on the distance to the forest trails.

In this article, we make the following two major contributions to counter the imbalance in calving front segmentation: 1) the use of MCC as an early stopping criterion, which is superior to a BCE monitored network, especially in the case of highly imbalanced data, and 2) further enhancing the performance of the network by using an improved distance map-based BCE loss. The former is compared with the BCE monitoring used in the state-of-the-art method and the latter is compared with the traditional BCE loss in one variant and with a weighted BCE loss in another variant. The experiments have been conducted on circa 400 SAR images from two glaciers. Comparing the performance metrics, i. e., intersection over union (IOU) and Dice coefficient, indicates the performance advantage of using MCC as the early stopping metric over BCE and using the improved distance map-based BCE over the traditional BCE loss.

The rest of this article is organized as follows. Section II reviews the tools that have been used in our contributions, i. e., MCC and the distance map loss. Section III describes our proposed pipeline and explains the main contributions of this article in detail. The dataset is explained in Section IV, while the experimental setup and the quantitative and qualitative

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results are presented and discussed in Section V. Finally, Section VI concludes this article.

II. Theoretical Background

A. Mathews Correlation Coefficient

BCE is a popular loss function in binary classification scenarios. A good loss is usually supposed not to produce very small or extremely large gradients, which is one of the desirable properties of the BCE loss [32]. We could choose to use the validation BCE loss also as an early stopping criterion. However, BCE loss is very fragile in the presence of class imbalance [33]. Hence, it becomes an unfavorable early stopping criterion for class-imbalanced segmentation.

Conversely, MCC [34] represents true positives (TPs), false negatives (FNs), false positives (FPs), and true negatives (TNs) in a balanced manner [35]. It is balanced in the sense that the rate of positive class predictions and negative class predictions is given equal importance. MCC is defined as

$$\mathrm{MCC} = \frac{\mathrm{TP} \times \mathrm{TN} - \mathrm{FP} \times \mathrm{FN}}{\sqrt{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}}. \quad (1)$$

It lies in the range $[-1, 1]$ with 1 representing the perfect prediction, -1 denoting wrong, and 0 suggesting random prediction. MCC takes positives and negatives equally into account and provides very little "discrimination" to random acts [23].

B. Distance Map Loss

A common approach to tackle the class imbalance with BCE loss is to weight each class with its inverse class frequency. However, it is not guaranteed to produce better performance [28]. Minority classes with higher weights can lead to noisy gradients [31]. On the contrary, a distance map-based approach adds context to the loss functions based on the regions of importance in the segmentation task. Distance maps, also known as distance fields or distance transforms, are derived representations of an image based on a distance metric.

A distance map-based loss trains the network by penalizing the predictions with respect to its distance from the object of interest in the image. A distance map-based loss is also beneficial for class imbalance. By forcing the network to give more attention to the regions of interest (minority class), it counters the bias of the network's training toward the majority class. The distance map loss works on the idea of weighting far away pixels (from the semantic line) lower than the ones in its proximity [31]. The loss is formulated as

$$J_{\mathrm{DW}} = \frac{1}{M \cdot N} \sum_{j=1}^{N} \sum_{i=1}^{M} \big((1 - g_{i,j}) \cdot d_{i,j} \cdot p_{i,j} - g_{i,j} \cdot d_{\max} \cdot \log(p_{i,j}) \big) \quad (2)$$

with g being the ground truth map and p the prediction at pixel location i, j. The distance map is denoted as d, with $d_{\max}$ being the maximum distance, and M and N represent the spatial dimensions of the maps.

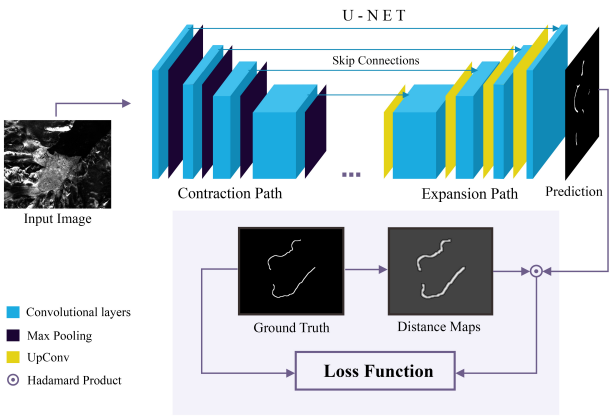

Fig. 1. Proposed glacier front segmentation pipeline.

III. Methodology

A. Processing Pipeline

Fig. 1 outlines our proposed segmentation workflow. We perform the segmentation of calving fronts images using a U-Net architecture, which was originally proposed by Ronneberger *et al.* [12]. The SAR images are used as input to the network, which learns either to predict the zones of ice mass regions or the calving front lines, depending on the ground truth data presented to the model at training time. In the rest of this article, we refer to the ice/nonice regions as zones and calving front locations as front lines for convenience.

The images are first preprocessed to remove noise, and then, patches of a predefined size were extracted from both the images and their corresponding ground truth before they are fed to the network. Before extracting the patches, the images are padded with zeros to the next larger image size that is a multiple of the patch size. The predicted masks on the patches are later stitched together to reconstruct the original size image, and the padding is discarded. In addition, in the variant that the network is trained to predict the ice zones, the predictions are further postprocessed to extract the boundaries of the predicted zones. The postprocessing is carried out in a two-step process as follows.

1) The wrongly predicted chunks of zones (the FPs) and the holes inside the zones (the FNs) are removed. The predicted zone regions are first sorted by area using the connected component analysis (CCA) [36]. All the connected components are filled with the maximum pixel value using a contour filling algorithm [37]. This removes all the FNs. Finally, we keep only one connected component with the largest area and discard the rest, and thereby remove all FPs.
2) To find the boundaries of the selected connected component from the previous step, we perform morphological erosion (ϵ_w) on the zones and subtract the eroded zones from the original zones. It is formulated as

$$L = Z - \epsilon_w(Z) \quad (3)$$

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where L denotes the boundary-extracted lines, Z denotes the predicted zones after connected component selection, and w denotes the kernel size of the morphological erosion.

B. MCC as the Early Stopping Metric

The data are split into three subsets, namely training, validation, and test sets. We use early stopping [21] to stop the training before it overfits. However, the validation error is usually very noisy. Hence, to ensure that the training terminates where the validation performance is optimum, we predefine a patience number. This is the number of epochs the early stopper waits to check for improvement on the validation error before terminating the training when no improvement is noticed. The common practice is to monitor the BCE loss on the validation set. However, due to the severe class imbalance, we propose the use of the MCC [34] as an early stopping metric as it is theoretically more robust toward class imbalance.

C. Modified Distance Map and Distance Map BCE Loss

We formulate the distance map in a generalized form to be used in different loss functions. The computed distance map weighs the line locations higher, while the regions around the proximity of the front lines are weighed less based on the distance to the lines. As we move progressively away from the front line locations, the weights keep reducing until they become 0. In other words, true line locations are given high importance and the proximity adds relaxation to the network. The points beyond which these weights become zero are considered to be the background (majority class). In contrast to the distance map-based approach by Adhikari and Kim [31], cf. (2), we weight the background differently depending on the class imbalance. In particular, we calculate the distance map from the ground truth masks as follows.

First, we thicken the ground truth lines by morphological dilation (δ_w) [38]. Then, we use the Euclidean distance transform (EDT), divided by a constant factor of R. The EDT keeps the pixels of the most central regions of the dilated lines as it is, but reduces the pixel intensity of the pixels that are far from it in proportion to its distance. The strength of relaxation is determined by R, i.e., how smoothly the weights fall off from the front lines to the background. The morphological dilation of the ground truth images thickens the front lines based on the structuring element used. We use a square structuring element with a size of (w, w), where w determines the amount of thickening of the lines. The result is further passed through a sigmoid function σ to obtain values in the range of $[0, 1]$

$$y_{\mathrm{dl}} = \delta_w(y) \tag{4}$$

$$y_{\mathrm{edt}} = \sigma\left(\frac{\mathrm{EDT}(y_{\mathrm{dl}})}{R}\right) \tag{5}$$

where y denotes the ground truth. Finally, the distance map d is obtained by

$$d = y_{\mathrm{edt}} + k(1 - y_{\mathrm{dl}}) \tag{6}$$

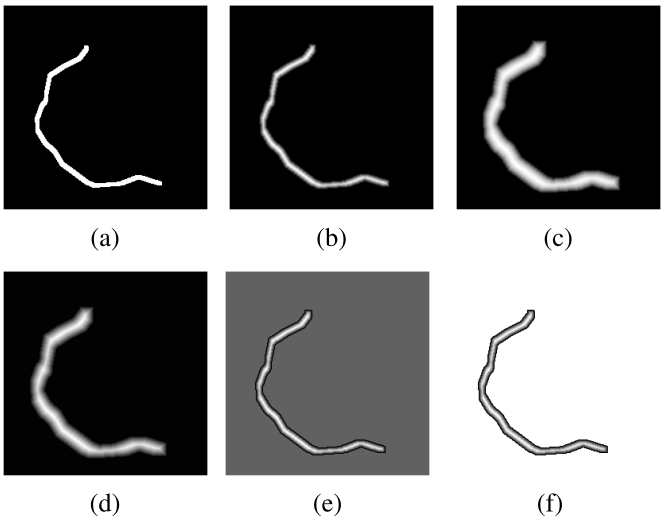

Fig. 2. Variation of the distance maps with different parameters when applied on the ground truth. (a) Ground truth. Distance maps with (b) $w = 5$, $R = 4$, and $k = 0.1$, (c) $w = 15$, $R = 4$, and $k = 0.1$, (d) $w = 15$, $R = 8$, and $k = 0.1$, (e) $w = 5$, $R = 4$, and $k = 0.5$, and (f) $w = 5$, $R = 4$, and $k = 1$. Here, $k = 1$ corresponds to white background and $k = 0$ corresponds to black background.

where k is a constant in the range of $(0, 1]$ determining the weight given to the background, with $k = 1$ signifying utmost importance. Since we deal with high class imbalance, a high weight is not desirable. However, it is a hyperparameter that depends on the dataset and its class imbalance ratio and needs to be tuned. Various distance maps are depicted with different values of w, R, and k in Fig. 2. Finally, the distance map computed on the ground truth is used in the BCE loss to create the distance map BCE loss. The BCE loss (J_{BCE}) is given by

$$J_{\mathrm{BCE}}(y, \hat{y}) = -\frac{1}{N}\sum_{i=1}^{N}\left(y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i)\right) \tag{7}$$

where $\hat{y}$ is the prediction, y is the ground truth, and N is the batch size during training. Incorporating the distance map into the BCE loss (J_{DBCE}) results in

$$J_{\mathrm{DBCE}}(y, \hat{y}) = -\frac{1}{N}\sum_{i=1}^{N}\big(y_i \log(\hat{y}_i \cdot d_i) + (1 - y_i)\log(1 - \hat{y}_i) \cdot d_i\big). \tag{8}$$

IV. Dataset

We perform our experiments on SAR images obtained from sites of the Antarctic Peninsula (AP) and Greenland, as shown in Fig. 3. AP has been strongly affected by changing climatic conditions [39]. The Dinsmoor–Bombardier–Edgworth (DBE) [40] and the Sjögren-Inlet (SI) [41] glacier had undergone a strong frontal retreat after the collapse of the downstream ice shelves in 1995 [40], [42]. Jakobshavn Isbræ [43] is one of the largest tidewater glaciers in Greenland and went through a drastic retreat over the last decades [44].

The AP is frequently covered by clouds, and therefore, time series of SAR satellite acquisitions provide the ideal data source compared to optical imagery. Data from the satellite

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TABLE I

OVERVIEW OF THE SAR SENSORS AND SPECIFICATIONS USED TO ACQUIRE THE SAR IMAGES OF THE AP AND JAKOBSHAVN ISBREA DATASETS

Platform	Sensor	Mode	SAR band	Repetition cycle [d]	Time interval	Multi looking factor	Ground range resolution [m]	Number of images	Train/Val/Test split
Antarctic Peninsula									
ERS-1/2	SAR	IM	C band	35/1	13. November 1995 - 26. February 2010	1×5	20	31	
RADARSAT 1 (SLC & PRI format)	SAR	ST	C band	24	10. September 2000 - 20. January 2008	1×4	20 (SLC) 12.5 (PRI)	49	
Envisat	ASAR	IM	C band	35	05. December 2003 - 03. July 2010	1×5	20	52	144/50/50
ALOS	PALSAR	FBS	L band	46	18. May 2006 - 03. March 2011	2×5	16,7	27	
TerraSAR-X TanDEM-X (SLC format)	SAR	SM	X band	11	13. October 2008 - 22. December 2014	3×3	6.7	85	
Jakobshavn Isbrea									
TerraSAR-X (EEC format)	SAR	SM	X band	11	16. April 2009 - 23. December 2015	5×5	6	149	99/25/25

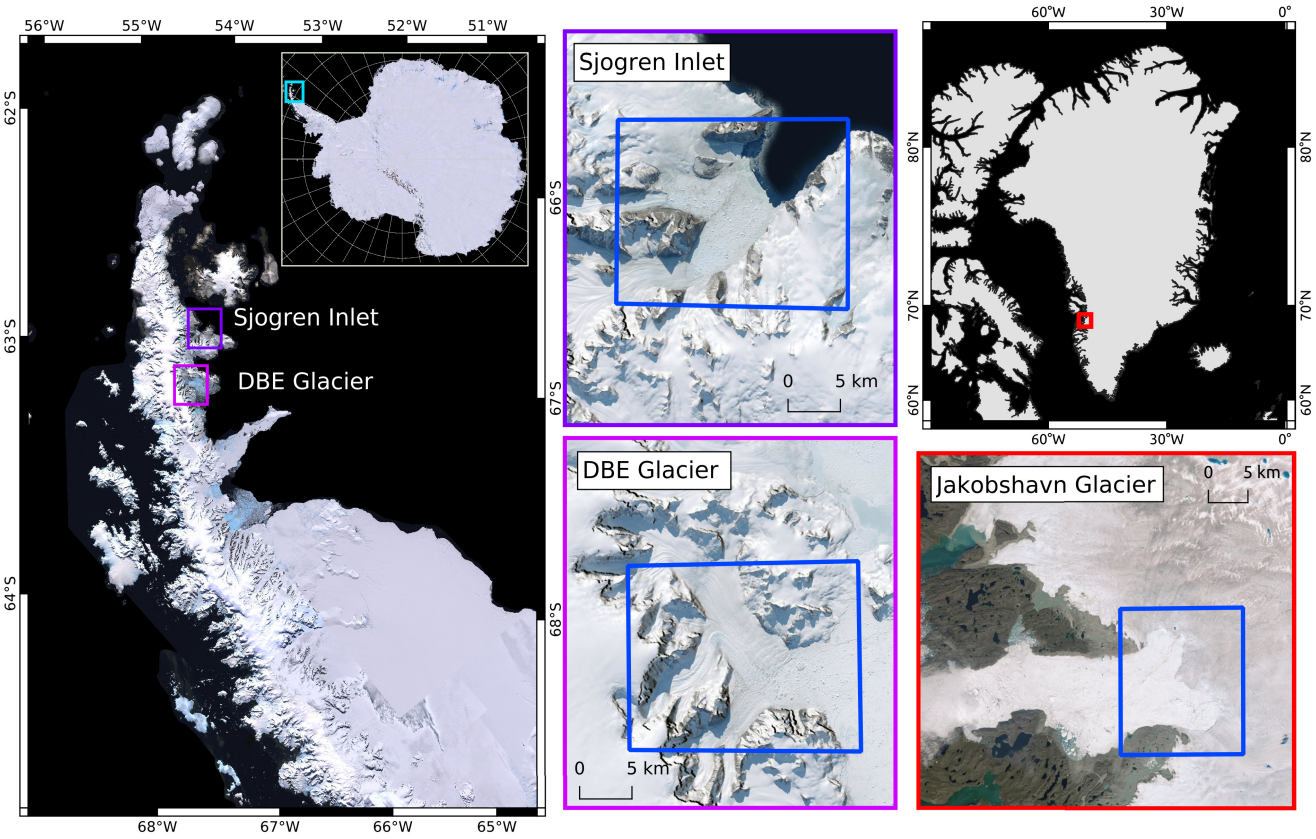

Fig. 3. Maps of the study regions and the locations of the studied glaciers at (Left) Northern AP and (Right) Greenland Jakobshavn. Blue polygons indicate subsets used for the CNN analysis. Background left panel: Landsat LIMA Mosaic © USGS, NASA, BAS, NFS. Background glacier subsets: ESRI Satellite © ESRI.

missions ERS-1/2, Envisat, RadarSAT-1, ALOS, TSX, and TanDEM-X (TDX) are analyzed, covering the period 199×–201×. The SAR images of the AP were sigma-0 calibrated and multilooked to remove speckle noise. Furthermore, the images were geocoded and orthorectified by means of the enhanced ASTER digital elevation model of the AP [45]. The ground truth of the front lines is taken from [40], [42]. The images vary on the basis of their spatial resolution. High spatial

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resolution images cover an area of 5 m × 5 m per pixel and range until the lowest resolution of 50 m × 50 m per pixel. Out of 244 available SAR images at the AP, randomly 50 images were kept for testing, another 50 were used for validation, and the rest served for training the model. We increase the AP training dataset three times by augmenting with horizontally flipped image versions as well as by 90° rotated versions. We refrain here from more augmentation variations to save computational time.

The Jakobshavn Isbræ images and glacier calving front locations were provided by Zhang *et al.* [14]. In particular, we obtained the relevant TSX enhanced-ellipsoid-corrected (EEC) products and applied the same preprocessing as Zhang *et al.* [14]. The authors kindly provided the preprocessing routines in order to obtain comparable input data for our analysis and to facilitate a comparison of the different approaches [14]. The SAR imagery used within this study was provided by the German Aerospace Center (DLR), the European Space Agency (ESA), and Alaska Satellite Facility (ASF). The SAR images of Jakobshavn Isbræ were preprocessed with a median filter [46] to remove speckle noise. All the images have a constant spatial resolution of 6 m × 6 m per pixel. We use 25 random images for validation, 25 for testing, and the remaining 119 images as the training set. The training data are augmented by adding horizontally flipped image versions. Further details on the dataset are shown in Table I. We have also added detailed information of the SAR data in the Supplementary Material.

At Jakobshavn Isbrae, SAR layover or shadow is not an issue, and hence, high-resolution data from one sensor (TSX) is used, whereas at the AP, we used multimission data (often much lower resolution), and due to pronounced topography at the AP leading to SAR layover and shadow, there are considerable no-data areas limiting the data availability.

The SAR images of AP also vary in size. The smallest of the images has dimensions of 377 × 458, and the largest one contains 3770 × 4581 pixels. Conversely, Jakobshavn Isbræ images are all approximately 1600 × 3500 pixels in size. For both datasets, two ground truth images are available: the calving front lines and the ice/nonice zones. Fig. 4 shows a sample SAR image in our dataset, along with its corresponding ice/nonice zones ground truth and its glacier calving front ground truth images. The zones have a training class imbalance of 1:3 in both datasets. The front line ground truth is severely imbalanced with a ratio of 1:2097 in AP training data and a ratio of 1:1124 in Jakobshavn Isbræ training data.

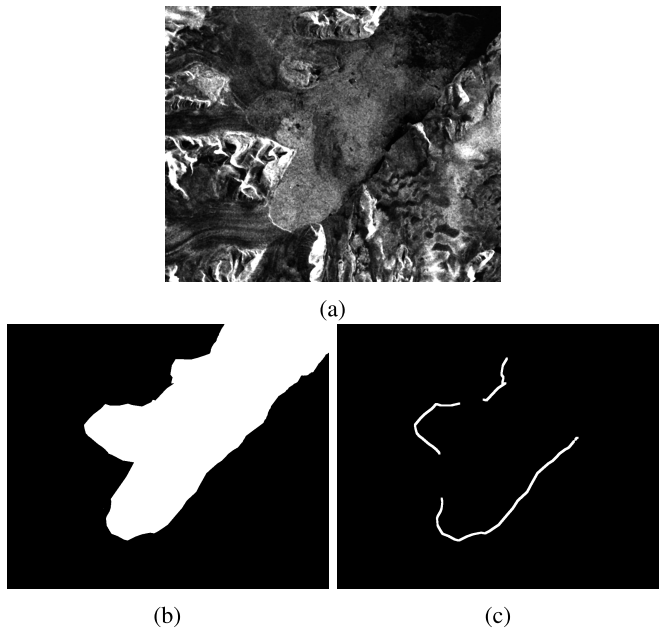

Fig. 4. (a) Example of an SAR image (AP glaciers) with its corresponding ground truth images. (b) Ice/nonice zones in which black represents the ice and white represents the nonice zones. (c) Glacier calving front line ground truth.

V. Experimental Setup and Results

A. Experimental Setup

1) Architecture and Parameters: We followed Zhang *et al.* [14] as our baseline architecture. Both the convolution and transposed convolution layers have kernel sizes of $(5, 5)$. We use leaky rectified linear units (ReLUs) as the activation functions with a negative slope of $\alpha = 0.1$, and the pooling layers have a pooling size of $(2, 2)$. For regularization, batch normalization layers have been exploited after every convolutional layer. Furthermore, we use Adam as the optimizer [47] and a cyclic learning rate for optimization, which is beneficial to the learning of the network [48]. Due to the constant changes in the network's performance by the cyclic training during training, we set a rather high patience value (30 epochs) for early stopping. Note that we also tried various other patience values for a subset of the experiments and the same behavior has been observed. Hence, we decided to use 30 for all experiments so that we give BCE early stopping criterion enough time to bring the validation loss down, too. We use a patch size of 256 × 256 pixels for patch extraction. The prediction of zones differs from the prediction of lines in the context of class imbalance. Hence, we use slightly different parameters for these two different ground truths, which are explained in the following experiments.

2) Experiment 1: Early Stopping Criterion: MCC Versus BCE: To demonstrate the importance of the metric used as the early stopping criterion, we perform the segmentation on the zones and the lines and compare the performance of BCE and MCC as the early stopping criteria. This experiment directly compares our method with the methodology that was proposed by Zhang *et al.* [14], where BCE was used as early stopping. For segmenting zones, a batch size of 20 is used. The maximum learning rate of $1e^{-2}$ and a minimum of $1e^{-7}$ is used in the cyclic learning rate with a step size of $5\times$ epochs.

In the second variant, where we train our model using the calving front lines directly, the front lines are one-pixel wide. This contributes to the major reason for the extreme class imbalance. To facilitate the network's training on such a highly imbalanced dataset, the lines were thickened using morphological dilation with a rectangular structuring element of size 5 × 5 pixels. This reduces the class imbalance of AP dataset lines to 1:420 and of Jakobshavn Isbræ to 1:225.

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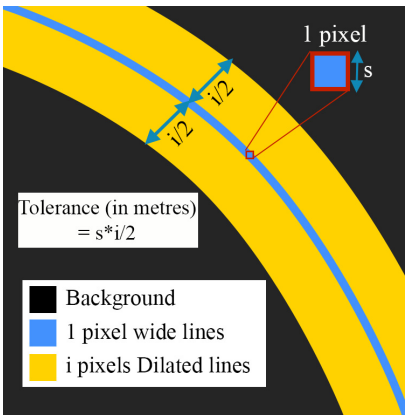
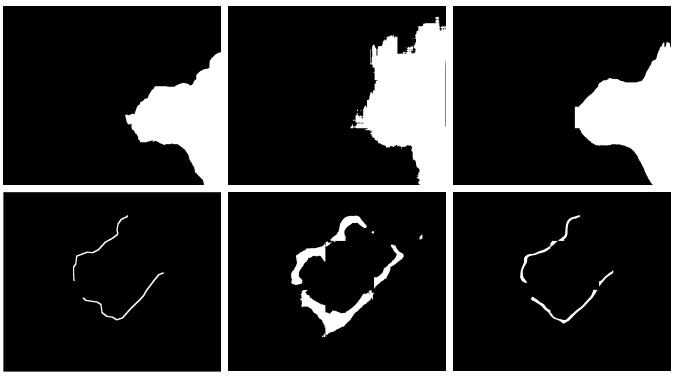


Fig. 5. Tolerance scheme based on dilation of lines and the spatial resolution of image.



(a) (b) (c)

Fig. 6. Comparing the segmentation results using BCE and MCC as the early stopping criterion on AP glaciers: (a) ground truth, (b) segmentation maps predicted using BCE as the early stopping criterion, and (c) segmentation maps predicted using MCC as the early stopping criterion. The top and bottom rows are the results on zones and lines, respectively.

We use a batch size of 15 for the front lines. The cyclic learning rates parameters for the lines are $1e^{-4}$ and $1e^{-8}$ as maximum and minimum, respectively.

3) Experiment 2: Improved Distance Map BCE Loss Versus BCE Loss: After proving the fact that MCC as an early stopping criterion accomplishes better results, we perform our second experiment using MCC as an early stopping criterion. In the second experiment, we compare the performance of the improved distance map BCE loss (i.e., with optimum k) with the conventional BCE loss, inverse class-frequency weighted BCE, and the conventional distance map BCE loss (i.e., with $k = 1$). This experiment is designed only to tackle high class imbalance. Hence, the models are trained only using the front line ground truth images of both datasets. We use the same parameters for predicting front line masks as in our first experiment. We search for the best parameters for the modified distance map formulation using grid search and, thereby, select $w = 3$ and $R = 1$ for both datasets. The best value of k was found to be 0.1 for AP and 0.25 for Jakobshavn Isbræ.

TABLE II
PERFORMANCE WHEN USING BCE AND MCC AS EARLY STOPPING CRITERION

Datasets	Metrics	Tolerance	Early Stopping Criterion	
			BCE	MCC
AP zones	IOU		0.8264	**0.8912**
	Dice Coeff		0.8984	**0.9389**
	MCC		0.8764	**0.9254**
	Number of training epochs		46	138
AP lines	IOU	65 m	0.1524	**0.2817**
		110 m	0.2201	**0.3957**
		155 m	0.2712	**0.4561**
	Dice Coeff	65 m	0.2617	**0.4421**
		110 m	0.3549	**0.5385**
		155 m	0.4196	**0.5997**
	MCC	65 m	0.3328	**0.4496**
		110 m	0.4112	**0.5456**
		155 m	0.4657	**0.6062**
	Number of training epochs		64	105
Jakobshavn Isbrae zones	IOU		0.9613	**0.9666**
	Dice Coeff		0.9801	**0.9828**
	MCC		0.9725	**0.9763**
	Number of training epochs		74	165
Jakobshavn Isbrae lines	IOU	60 m	0.3755	**0.5681**
		105 m	0.4424	**0.6451**
		150 m	0.4801	**0.6928**
	Dice Coeff	60 m	0.5437	**0.7223**
		105 m	0.6108	**0.7824**
		150 m	0.6459	**0.8167**
	MCC	60 m	0.5712	**0.7208**
		105 m	0.6273	**0.7783**
		150 m	0.6553	**0.8103**
	Number of training epochs		43	106

B. Evaluation

1) Evaluation Metrics: We use the performance metrics that are commonly used for segmentation problems [9], [49], i.e., IOU and Dice coefficient, to evaluate our experiments. We also report the MCC. IOU is the area of overlap between the predicted masks p and the ground truth masks g divided by the area of union between the predicted masks and the ground truth masks. Here, the ground truth masks refer to data based on expert interpretation of remote sensing data. The Dice coefficient is twice the area of overlap divided by the total number of pixels from predictions and ground truth

$$\text{IOU} = \frac{p \cap g}{p \cup g} \tag{9}$$

$$\text{Dice} = \frac{2\sum_{i=1}^{N} p_i g_i}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} g_i}. \tag{10}$$

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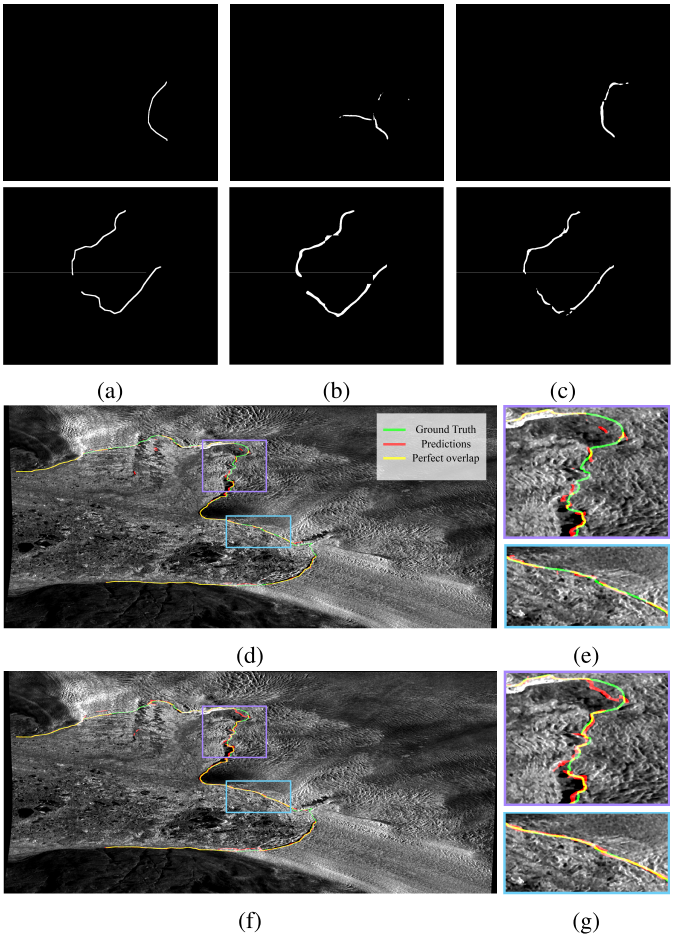

Fig. 7. (a) Two ground truth front line samples of the AP glacier. Predicted lines achieved by (b) BCE loss and (c) distance map BCE loss. Comparison of predictions of (d) BCE loss and (f) DMap BCE loss on Jakobshavn Isbrae glaciers. (e) Magnified versions of the marked areas from the BCE Loss prediction image. (g) Magnified areas from the DMap BCE loss prediction image.

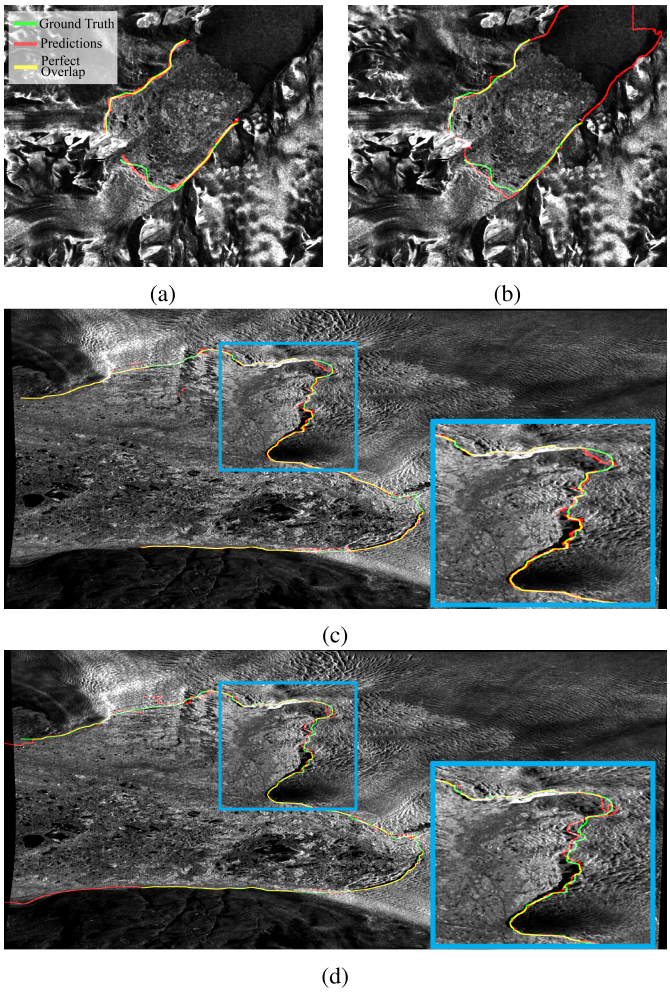

Fig. 8. Comparison of the predictions achieved by (a) front lines model and (b) boundary extracted from zones model on the AP glaciers. Comparison of the predictions achieved by (c) front lines model and (d) boundary extracted from zones model on the Jakobshavn Isbrae glaciers.

These performance metrics solely represent how well the predictions overlap with the ground truth masks. This makes it hard to evaluate the calving front lines, because even if they are very close to the ground truth locations, they will be considered as a miss by these metrics. Therefore, to make these metrics more comprehensible, we further dilate the predictions and ground truth masks. The dilated pixels around the ground truth are considered as the relaxation or tolerance. The tolerance is calculated based on the spatial resolution of our datasets, which is 15 m (average) and 6 m for AP and Jakobshavn Isbrae, respectively. The spatial resolution s denotes the actual distance one pixel covers on Earth. Therefore, the tolerance t is calculated as $t = s \times i/2$, where i denotes the average pixel width of the dilated lines. The tolerance scheme is shown in Fig. 5. The metrics are reported in our tables along with the tolerance (in meters) for the predicted calving front lines.

2) Experiment 1: Early Stopping Criterion: MCC Versus BCE: Table II shows the results when using BCE and MCC as an early stopping criterion. Monitoring MCC clearly outperforms BCE in all cases. Prediction of zones in Jakobshavn Isbræ does not show much improvement as BCE as early stopping performed very well, having IOU of 96.13% and Dice score of 98.01%. The reason is that the use of MCC continues training to roughly 150 epochs, while BCE already stops at around 50 epochs. Validation BCE decides that the network is overfitting too early. On the other hand, MCC is more comprehensive about the validation error and correctly allows the network to be trained further. Table II also shows that monitoring MCC shows improvement not only when dealing with high class imbalance but also is quite reliable for less imbalanced datasets (AP and Jakobshavn Isbræ zones). The fact that the BCE-monitored network is not trained enough can be clearly observed in the qualitative results that are shown in Fig. 6 showing mosaicing artifacts and oversegmentation. The MCC-monitored network performs significantly better for both zones and lines. Apart from the fact that the calving front lines generated from the MCC monitored network are more accurate, the predictions also seem to have fewer numbers of FPs (incorrect white lines on the black regions), as shown in Fig. 6.

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TABLE III

PERFORMANCE OF THE IMPROVED DISTANCE MAP BCE LOSS (I. E., WITH OPTIMUM k) VERSUS THE BCE LOSS, INVERSE CLASS-FREQUENCY WEIGHTED BCE, AND THE CONVENTIONAL DISTANCE MAP BCE LOSS

Datasets	Metrics	Tolerance (meters)	Loss function			
			BCE	BCE (with weights)	Dmap BCE, $k = 1$	Dmap BCE, optimum k
Antarctic Peninsula lines	IOU	65	0.2817	0.2798	0.3078	**0.3085**
		110	0.3957	0.3728	0.3991	**0.4072**
		155	0.4561	0.4355	0.4591	**0.4734**
	Dice Coeff.	65	0.4421	0.4211	0.4456	**0.4463**
		110	0.5385	0.5242	0.5442	**0.5506**
		155	0.5997	0.5873	0.6046	**0.6152**
	MCC	65	0.4496	0.4302	0.4502	**0.4544**
		110	0.5456	0.5295	0.5467	**0.5566**
		155	0.6062	0.5913	0.6059	**0.6204**
Jakobshavn Isbrae lines	IOU	60	0.5681	0.5535	0.5675	**0.5908**
		105	0.6451	0.6272	0.6417	**0.6714**
		150	0.6928	0.6717	0.6872	**0.7182**
	Dice Coeff.	60	0.7223	0.7101	0.7217	**0.7401**
		105	0.7824	0.7684	0.7794	**0.801**
		150	0.8167	0.8012	0.8123	**0.8338**
	MCC	60	0.7208	0.707	0.7204	**0.7364**
		105	0.7783	0.7031	0.7754	**0.7957**
		150	0.8103	0.7937	0.8058	**0.8271**

TABLE IV

PERFORMANCE OF THE BOUNDARY-EXTRACTED ZONES VERSUS DIRECT FRONT LINE PREDICTIONS OF THE CALVING FRONT LOCATION

Dataset	Metrics	Tolerance (Meters)	Model		
			Boundary-Extracted zones	Rock-coastline removed from boundary extracted zones	Direct front lines
Antartic Peninsula	IOU	65 m	0.0931	0.2003	**0.3085**
		110 m	0.1471	0.3112	**0.4072**
		155 m	0.1858	0.3894	**0.4734**
	Dice Coeff	65 m	0.1661	0.309	**0.4463**
		110 m	0.2504	0.4466	**0.5506**
		155 m	0.3065	0.5338	**0.6152**
	MCC	65 m	0.1871	0.3089	**0.4544**
		110 m	0.2768	0.4452	**0.5566**
		155 m	0.3351	0.5316	**0.6204**
Jakobshavn Isbrae	IOU	60 m	0.4705	0.5266	**0.5908**
		105 m	0.5753	0.65	**0.6714**
		150 m	0.6348	**0.7204**	**0.7182**
	Dice Coeff	60 m	0.6358	0.6858	**0.7401**
		105 m	0.7267	0.7842	**0.801**
		150 m	0.7732	**0.8342**	**0.8338**
	MCC	60 m	0.6307	0.6799	**0.7364**
		105 m	0.718	0.777	**0.7957**
		150 m	0.7646	**0.8262**	**0.8271**

3) Experiment 2: Improved Distance Map BCE Loss Versus BCE Loss: The results of BCE loss and improved distance map BCE loss are shown in Table III. The improved distance map BCE loss is also compared with the distance map using ($k = 1$). It is interesting to notice that the distance map BCE loss with $k = 1$ did not improve the performance of BCE loss in our task of glacier front line segmentation. However, it does show slight improvement with low tolerance (65 m) for the AP glaciers. In contrast, the optimum k helps to deal with the class imbalance better. The qualitative results of the predicted lines on Jakobshavn Isbræ regions are shown in Fig. 7. In this figure, the green color represents ground truth, the red color shows the incorrect prediction, and the yellow color depicts the correct predictions, i. e., the overlap between the prediction and the ground truth. The line predictions of Jakobshavn Isbræ glaciers in both losses are disconnected. However, the distance map

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loss manages to identify more regions of the calving fronts. BCE loss also incorrectly predicts severe outliers (FPs) that are far away from the real locations, as shown in Fig. 7. Such FPs are less numerous in the case of the distance map BCE predictions. Referring to Fig. 7(a)–(c), which depicts the front line predictions of the AP glaciers, we see that the BCE loss predicts several images quite inaccurately. These images are well handled by the distance map loss.

Note that the performance on the AP glaciers is lower than the Jakobshavn Isbrae glaciers for both the experiments (see Sections V-B2 and V-B3). This is presumably due to differences between the two datasets. The AP glaciers vary a lot in terms of image size, ground range resolution, and also with multilooking factors as shown in Table I, while Jakobshavn has the same sensor acquisition, constant large image size, and high resolution.

4) Comparison of Zone and Line Networks: When working on the problem of automatic detection of CFPs in the SAR images, the available ground truth data for the training could be the ice zones or the calving front lines. The semantic segmentation of zones on the SAR images needs postprocessing to find the calving front locations. Unlike semantically segmenting the zones, a direct semantic segmentation of the calving front locations is a harder problem due to the more severe class imbalance. Fig. 8 qualitatively compares the segmented front lines using the two approaches for the AP glaciers dataset: one approach predicts the front lines directly and the other postprocesses the predicted zones. Direct predictions of calving front lines bring minor disjoints in the predictions, whereas the postprocessed zone predictions are continuous. However, the result of the boundary-extracted zone is not entirely the glacier calving front lines. It shows the boundary between ice zones and nonice zones, which may contain some rock-coastline boundaries as well. The rock-coastline boundaries also being predicted from the postprocessed zone predictions can be seen in Fig. 8 as the extra red lines. To have more accurate calving front lines derived from boundary-extracted zones, further postprocessing steps are needed, e. g., a manual removal of rock coastline boundaries or an automatic way that may require additional prior knowledge. However, apart from the disjoints, the regions that are predicted by direct front line predictions are more accurate compared to the boundary-extracted zones, as shown in Fig. 8.

The difference in the performance of the boundary-extracted zones and the lines for predicting calving front location is compared quantitatively in Table IV. The table also quantizes the glacier calving front extraction from the zones before and after manually removing the extra FPs (rock-coastline boundaries). We see that the model that is trained to predict the lines directly outperforms the ones that segment the zones. However, with high tolerance, the predicted calving front locations from the boundary-extracted zones are comparable with the lines model for the Jakobshavn Isbrae glaciers. This is because, unlike AP glaciers, the zone predictions of the Jakobshavn Isbrae were highly accurate (as shown in Table II), and with increasing tolerance, more predicted regions tend to overlap with the ground truth.

Apart from the differences in performance, they also differ in their processing pipeline. In comparison to direct front line model, extracting boundaries from the zones has an additional hyperparameter (kernel size of morphological erosion) that needs to be tuned. The manual postprocessing of removing the FPs (rock-coastline boundaries) prevents the overall pipeline of boundary-extracted zones to be end-to-end automatic. On the other hand, the direct front lines model is not only end-to-end automatic and faster but also has a higher training speed (reported in Table II).

VI. Conclusion

This article proposes methods to improve deep learning-based semantic segmentation of the glacier calving fronts. The detection of calving fronts is challenged by its severe class inequality. We have tackled the class-imbalance problem by two approaches. First, the use of MCC as an early stopping criterion showed a great performance gain. Then, to help the BCE loss function to handle the imbalanced data, we incorporated a novel distance map loss. The distance map can be used in conjunction with any other loss function. The MCC-monitored network outperforms the BCE-monitored network by 2.16% dice coefficient for the zone predictions and by 17.43% for the line predictions. Using the modified distance map BCE loss results in a further improvement of the line predictions by 1.57% dice coefficient.

Finally, two approaches of predicting front lines were compared, i. e., either to directly predict the front lines or to predict ice zones and then postprocess the areas to delineate the calving fronts. The good quality of the results highlights the suitability of the tested approach for glacier front detection on SAR images. Therefore, the next steps could include applying the system to other glaciers and on larger areas, with the goal of developing an automated monitoring system of calving glaciers. The presented results in this work are encouraging and we believe that other image segmentation applications with severe class imbalance, such as supra-glacial lake detection, road extraction, ship detection, or others, can also benefit from the proposed approaches in this work.

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Implementation

Our codes, which we have used to produce the experimental results in this work, are available through GitHub [50].

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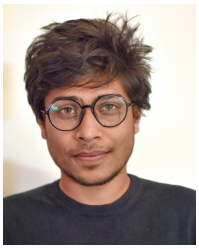

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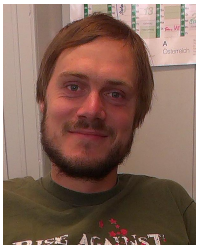

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